MERCON Logistics — Product Blueprint

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1. Executive Summary

The MERCON Logistics Management System is a comprehensive digital logistics platform designed to streamline and modernize the transportation operations of MERCON Logistics Services Company. The platform replaces fragmented manual processes — including phone calls, messaging applications, spreadsheets, and paper-based records — with a centralized, cloud-connected ecosystem that enables efficient planning, execution, monitoring, and management of logistics operations.

The system is built around four integrated platforms:

Platform	Users	Purpose
Driver Operations Ap	Drivers	Execute assigned trips through a guided workflow with live tracking and digital delivery verification
Operator Mobile App	Operators	Create trips, monitor live operations, receive alerts, and perform quick actions while mobile
Operator Web Dashb	Operators	Full trip management, customer/driver/fleet management, pricing, live monitoring, and reporting
Admin Web Dashboa	Administrator	User & role management, fleet/customer admin, pricing, reports, analytics, integrations, and system settings

Together, these platforms provide seamless communication and real-time synchronization between field personnel and office operations, ensuring complete visibility throughout the transportation lifecycle.

Primary objectives: Improve operational efficiency, increase fleet utilization, enhance communication between operators and drivers, reduce manual administrative effort, and provide accurate real-time visibility across all transportation activities.

The first release (MVP) focuses on the core transportation lifecycle, including user authentication, trip management, route and pricing management, driver and vehicle assignment, live GPS tracking, pickup and delivery verification, document management, notifications (including WhatsApp trip sharing), operational reporting, and dashboard analytics. Invoicing is out of scope for MVP; only quotations are supported.

The platform has been designed with scalability in mind, allowing future versions to introduce advanced capabilities such as customer self-service portals, AI-assisted route optimization, predictive analytics, ERP integrations, automated billing & invoicing, offline mode with auto-sync, and additional enterprise management features.

2. Vision & Mission

Vision Statement

To become a leading technology-driven logistics management platform that empowers transportation companies to operate with greater efficiency, transparency, and reliability by digitizing every stage of the logistics lifecycle. MERCON aims to create a connected ecosystem where operators, drivers, and administrators collaborate seamlessly through real-time information, intelligent workflows, and centralized operations.

Mission Statement

To deliver a unified, secure, and scalable logistics management platform that simplifies transportation operations by connecting operators, drivers, and administrators through a centralized digital ecosystem. By replacing manual processes with intelligent workflows, real-time communication, and data-driven decision-making, the MERCON Logistics Management System enables organizations to manage trips, fleets, drivers, and operational activities with greater accuracy, efficiency, and transparency.

3. Product Goals

Business Goals

- Digitize and centralize all logistics operations within a single integrated platform
- Reduce dependency on manual communication methods
- Improve operational transparency with real-time visibility
- Increase fleet utilization through efficient allocation
- Improve customer satisfaction with timely, accurate updates
- Build a scalable platform for future growth

Operational Goals

- Standardize the transportation lifecycle end-to-end
- Enable operators to create, assign, and monitor requests from both web and mobile
- Provide drivers a simple app for assignments and progress
- Maintain accurate records for every trip
- Improve operator–driver communication in real time
- Automate routine processes and reporting

User Goals

Role	Goals
Drivers	Receive trip assignments instantly · Access all trip info in one place · Navigate efficiently with integrated maps · Complete pickup/delivery v
Operators	Create and manage requests efficiently (web and mobile) · Assign drivers and vehicles with minimal effort · Monitor live movement and trip
Administra	Manage users, vehicles, customers, pricing centrally · Monitor performance through analytics · Maintain secure role-based access control ·

Technical Goals

- Develop a secure, scalable, cloud-based logistics platform
- Support real-time synchronization between mobile apps and web dashboards

- Implement role-based authentication and secure access control
- Enable live GPS tracking (Mobile GPS + ICCES) and instant operational notifications
- Design a modular architecture supporting future expansion and integrations
- Ensure high availability, reliability, performance, and data security at scale

4. Target Users

Driver — Field Workforce

The Driver Operations App acts as a guided operational assistant, presenting only the actions required at each stage of the trip — from starting a trip to completing delivery verification.

Responsibilities: Start assigned trips · Capture cargo photos · Share live GPS location · Report emergencies · Capture delivery photos · Complete the trip

Challenges Today: Frequent phone calls in transit · Manual progress reporting · Delayed issue communication · Manual delivery evidence

How MERCON Helps: Automated stage-by-stage guidance · Live GPS & auto notifications · Digital proof of delivery · Secure Employee ID or Mobile Number / PIN login

Operator — Operational Control Center

Operators create trips, assign drivers and vehicles, monitor live vehicle movement, respond to incidents, and ensure successful delivery completion — from both web and mobile.

Responsibilities: Create transportation requests · Assign drivers & vehicles · Monitor active trips · Track via GPS & ICCES · Verify deliveries

Challenges Today: Monitoring multiple active trips · Coordinating dispersed drivers · Responding to fast-changing ops · Keeping accurate records

How MERCON Helps: Centralized real-time monitoring · Live GPS & emergency alerts · Automated trip updates · Delivery confirmations · Trip creation from mobile

Administrator — Platform & Business Oversight

Administrators focus on long-term system management and business oversight — maintaining organizational data, fleet, pricing, and platform security.

Responsibilities: Manage users & permissions · Maintain fleet information · Manage customer records · Configure pricing & settings · Manage integrations

Challenges Today: Growing operational data · Security & access control · Monitoring fleet performance · Ensuring data accuracy

How MERCON Helps: Centralized dashboard control · Analytics & reporting · Fleet & pricing oversight · Governed system configuration · Integration management

User Summary

User Role	Platform	Primary Objective
Driver	Driver Operations App	Execute assigned transportation trips through a guided workflow with live tracking and digital delivery v
Operator	Operator Mobile App & Operator W	Plan, assign, monitor, and manage transportation operations in real time
Administra	Admin Web Dashboard	Manage users, fleet, pricing, reports, analytics, integrations, and overall platform administration

5. Product Ecosystem

The MERCON Logistics Management System connects customers, operators, drivers, fleet resources, and administrators within a single digital ecosystem — streamlining the complete transportation lifecycle from customer request to successful delivery.

Platform Descriptions

Driver Operations App

A workflow-driven app that guides drivers through every stage of an assignment — cargo photos, live GPS, emergencies, delivery proof, and completion.

Operator Mobile App

Extends operational capability beyond the office — create trips, monitor live trips, receive notifications, track drivers, and respond to events while mobile.

Operator Web Dashboard

The primary control center — trip creation & management, quotations, driver & vehicle assignment, live monitoring, fleet and customer management, pricing, and reporting.

Admin Web Dashboard

Centralized administrative control — users, permissions, pricing, fleet records, customer records, reports, analytics, integrations, audit logs, and system configuration.

Ecosystem Architecture

```
CUSTOMER
Transportation Request — Daily / Monthly / Weekly / One-Time
        ↓
OPERATOR WEB DASHBOARD / OPERATOR MOBILE APP
Create Trip · Generate Quotation · Assign Driver & Truck · Auto Pricing
Customer Management · Fleet Monitoring · Share Trip (WhatsApp/PDF/Link) · Reports
        ↓
CENTRALIZED CLOUD PLATFORM
Authentication · Business Logic · Database · Pricing Engine · Notification Engine
Trip & Fleet Management · Reports · Analytics · Media Storage · GPS Processing
Role Management · Audit Logs · APIs
        ↓
┌──────────────────────┬───────────────────────┬─────────────────────────┐
│ Driver Operations    │ Operator Mobile App   │ Admin Web Dashboard     │
│ App                  │                       │                         │
│ • Start Trip         │ • Create Trip         │ • User Management       │
│ • Cargo Photos       │ • Live Trip Monitor   │ • Fleet Management      │
│ • Live GPS           │ • Notifications       │ • Pricing Management    │
│ • Emergency          │ • Emergency Alerts    │ • Customer Management   │
│ • Delivery Photos    │ • Driver Tracking     │ • Reports & Analytics   │
│ • Complete Trip      │ • Quick Actions       │ • Integrations          │
│                      │ • Trip Status         │ • Roles & Permissions   │
│                      │                       │ • Audit Logs            │
└──────────────────────┴───────────────────────┴─────────────────────────┘
        ↓
GOOGLE MAPS GPS (Primary Tracking)
ICCES GPS DEVICE (Secondary Truck Tracking)
        ↓
REAL-TIME LOCATION & STATUS UPDATES
        ↓
DELIVERY COMPLETED
Operator Notified → Reports Updated → Customer History Updated
```

6. Transportation Lifecycle

Step	Description	Status
1	Customer submits a transportation request	—
2	Operator creates trip record	Created
3	Pickup and delivery locations defined via Google Maps	Created
4	Cargo details and special instructions captured	Created
5	Trip scheduled (one-time, daily, weekly, or monthly)	Created
6	System suggests pricing based on route and customer rate cards	Created
7	Available, compliant vehicle assigned	Assigned
8	Available driver assigned; trip pushed to Driver App	Assigned
9	Trip shared to Driver & Customer (WhatsApp / PDF / Link)	Assigned
10	Driver opens and acknowledges trip	Ready to Start
11	Driver taps "Start Trip" → mandatory cargo photo upload	Cargo Verification
12	Trip officially begins; navigation opens automatically	In Transit

13	Continuous GPS location sync with Operator Dashboard	In Transit
14	Emergency handling if required (operator notified immediately)	Emergency / Halted
15	Arrival at destination detected via GPS geofencing (500m radius)	Arrived
16	Mandatory delivery photo upload (proof of delivery)	Delivery Verification
17	Trip completed; timestamp recorded automatically	Completed
18	Reports, analytics & customer history updated	Completed

7. Product Scope

In Scope — MVP

Authentication & User Management

- Secure Login (Email/Password for Admin/Operator; Employee ID or Mobile Number + PIN for Driver)
- Role-Based Access Control
- User Profiles
- Session Management

Driver Operations App

- Guided Trip Workflow
- Cargo & Delivery Photos
- Live GPS & Navigation
- Emergency Reporting

Operator Mobile App

- Trip Creation (full flow)
- Live Trip Monitoring
- Driver Tracking
- Emergency Alerts
- Quick Actions

Operator Web Dashboard

- Dashboard & KPIs
- Trip & Route Management
- Pricing Engine & Quotations
- Fleet, Driver & Customer Management
- Live Monitoring & Notifications
- Mobile GPS + ICCES Integration
- Trip Timeline
- Automated Alerts

Admin Web Dashboard

- User & Role Management
- Fleet & Customer Admin
- Pricing Management

- Reports & Analytics
- Audit Logs
- Integrations
- System Settings

Reports

- Daily / Weekly / Monthly
- Customer, Driver & Vehicle Reports
- Revenue Reports

Shared Platform Features

- Real-Time Sync & Push Notifications
- Media Upload & Cloud Database
- Google Maps + ICCES Integration
- WhatsApp / PDF / Link Trip Sharing

Out of Scope — MVP

- Customer Mobile Application
- Online Payment Gateway
- AI Route Optimization
- Driver Chat System
- Fuel Management
- Warehouse / Inventory Management
- ERP & Accounting Integrations
- Offline Mode with Auto-Sync
- Automated Billing & Invoicing

Future Enhancements

- Customer Self-Service Portal
- AI-Based Route Optimization
- Dynamic Pricing Engine
- Automated Billing & Invoicing
- Digital Signature for Delivery
- Driver Performance Scoring
- Fuel & Fleet Health Analytics
- Predictive Vehicle Maintenance
- IoT Sensor Monitoring
- Multi-language Support
- Offline Trip Synchronization with Auto-Sync
- Public Customer Tracking Portal
- Business Intelligence Dashboard
- ERP / CRM & 3rd-Party Integrations
- Multi-Company / Multi-Branch Operations

8. Core Modules

Sixteen interconnected functional modules manage the complete transportation lifecycle:

#	Module	Primary Purpose
01	Authentication & Access Management	Secure user authentication and role-based access
02	Dashboard	Operational overview and business insights
03	Trip Management	Complete transportation lifecycle management
04	Route & Navigation	Route planning and navigation
05	Pricing & Quotation	Transportation pricing and quotations
06	Cargo Management	Shipment information and cargo verification
07	Driver Management	Driver administration and assignment
08	Fleet Management	Vehicle and fleet administration
09	Customer Management	Customer records and business relationships
10	Live Tracking	Real-time trip and vehicle monitoring
11	Driver Workflow	Guided transportation execution
12	Notification & Communication	Real-time alerts, push notifications, WhatsApp sharing
13	Reports & Analytics	Operational reporting and performance analysis
14	Administration	Platform administration and configuration
15	Document Management	Secure document storage and compliance
16	Platform Services	Core backend infrastructure and shared services

9. Business Rules

BR-001 · User Authentication & Access Control

- Every user must authenticate using authorized credentials before accessing the system
- Admin & Operator: Email + Password
- Driver: Employee ID or Mobile Number + PIN
- Users shall only have access to modules permitted by their assigned role
- Sessions must automatically expire after prolonged inactivity
- All user activities shall be recorded in the audit log

BR-002 · Driver Assignment

- A driver may only have one active trip at any given time
- Only drivers marked as Available may be assigned to a trip
- Drivers cannot manually assign or modify trips
- Assignments become immediately available in the Driver App with a push notification

BR-003 · Vehicle Assignment

- A vehicle may only be assigned to one active trip at a time

- Vehicles under maintenance cannot be assigned
- Vehicles with expired documentation are unavailable for assignment
- Only vehicles with sufficient capacity may be assigned to the selected cargo

BR-004 · Trip Creation

- Every trip must be associated with a registered customer
- Pickup and delivery locations are mandatory
- Cargo information must be completed before a trip can be created
- A driver and vehicle must be assigned before status becomes Assigned
- Each trip shall have a unique Trip ID

BR-005 · Pricing & Quotations

- The system shall automatically suggest pricing based on predefined route rates
- Operators may manually override the suggested price
- Customer-specific pricing takes precedence over default pricing
- Every pricing modification shall be recorded

BR-006 · Driver Workflow

- If an active trip exists, the Driver App opens directly to the current trip
- Driver taps "Start Trip" to begin the pre-departure checklist
- Drivers cannot proceed without completing mandatory cargo photo verification
- Trip status changes to In Transit after cargo photos are successfully uploaded
- The navigation screen automatically opens after the trip begins

BR-007 · GPS Tracking

- Driver mobile GPS is the primary source of live location tracking
- ICCES vehicle GPS functions as a secondary tracking source when available
- GPS tracking begins automatically when a trip starts and ends when it completes
- Location updates synchronize continuously with the Operator Dashboard

BR-008 · Emergency Handling

- Drivers may report emergencies at any time during an active trip
- Emergency alerts shall immediately notify the assigned operator
- The driver's live GPS location shall accompany every emergency report
- Emergency events shall be stored within the trip history

BR-009 · Arrival & Delivery Verification

- The application shall detect arrival using GPS geofencing (default 500m radius)
- Drivers receive an automatic "You Have Arrived" notification
- Fallback: Manual "I Have Arrived" button if geofence doesn't trigger
- Delivery photographs are mandatory before completing a trip
- Trip status changes to Completed after successful delivery verification

BR-010 · Notifications

- Auto-generated for: Trip Assigned, Started, Cargo Uploaded, Driver Arrived
- Auto-generated for: Emergency Reported, Trip Halted, Delivery Uploaded, Completed
- Auto-generated for: Vehicle Maintenance & Document Expiry Reminders

BR-011 · Operator Operations

- Operators may create, edit, and manage transportation requests (web and mobile)
- Operators may assign or reassign drivers before a trip starts
- Operators receive real-time updates for every trip event
- Operators may generate operational reports at any time

BR-012 · Fleet Management

- Every truck shall have a unique fleet record
- Maintenance schedules shall be recorded for every vehicle
- Vehicle documentation shall include expiry dates
- Vehicles with expired compliance documents are unavailable for assignment

BR-013 · Customer Management

- Every customer shall have a unique customer profile
- Customer trip history is retained permanently unless archived
- Customer quotations are linked to transportation requests

BR-014 · Document Management

- Trip photographs are permanently linked to their corresponding trip
- Driver and vehicle documents include expiry tracking
- Document renewal reminders are generated automatically

BR-015 · Reporting

- Reports generated for Trips, Drivers, Fleet, Customers, Revenue, Pricing, and Performance
- Reports available daily, weekly, monthly, and for custom date ranges

BR-016 · Audit & Security

- All critical system actions shall be logged
- Role-based permissions shall be enforced across all applications
- Sensitive operational data shall be securely stored and transmitted
- Deleted records remain recoverable per the organization's retention policy

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10. Information Architecture

Driver Operations App

```
Driver Operations App
├── Splash Screen
├── Login (Employee ID or Mobile Number + PIN)
├── Waiting for Assignment
├── Active Trip
│   ├── Trip Summary
│   ├── Start Trip
│   ├── Cargo Photo Upload
│   ├── Navigation
│   ├── Emergency
│   ├── Arrival Detection
│   ├── Delivery Photo Upload
│   └── Complete Trip
├── Trip History
├── Notifications
├── Profile
└── Settings
```

Operator Mobile App

```
Operator Mobile App
├── Login (Email + Password)
├── Dashboard
│   ├── Live Trips
│   ├── Today's Trips
│   ├── Scheduled Trips
│   ├── Driver Availability
│   ├── Emergency Alerts
│   ├── Maintenance Alerts
│   └── Document Expiry
├── Trip Management
│   ├── Trip List
│   ├── Trip Details → Live Tracking → Live Map
│   └── Create Trip (Route → Cargo → Schedule → Pricing → Vehicle → Driver → Review)
├── Alerts
│   ├── Trip Alerts
│   ├── Emergency Alerts
│   ├── Document Expiry
│   ├── Maintenance Alerts
│   └── System Notifications
├── Profile
└── Settings
```

Operator Web Dashboard

```
Operator Dashboard
├── Dashboard
│   └── KPIs · Today's Trips · Live Activity · Fleet Status · Driver Availability · Notifications
├── Trip Management
│   └── Create · Active · Scheduled · Completed · Cancelled · Trip Details · Share Trip (WhatsApp/PDF/Link)
├── Live Monitoring
│   └── Live Map · Driver GPS · ICCES GPS · Trip Timeline · Emergency Alerts
├── Customer Management
│   └── Customers · Profile · Quotations · Contracts · Trip History
├── Driver Management
│   └── Drivers · Profile · Availability · Documents · Performance
├── Fleet Management
│   └── Trucks · Trailers · Maintenance · Documents · Availability
├── Pricing
│   └── Route Pricing · Customer Pricing · Quotations · Pricing History
├── Reports
│   └── Daily · Weekly · Monthly · Driver · Fleet · Customer · Revenue
├── Notifications
├── Profile
└── Settings
```

Admin Web Dashboard

```
Admin Dashboard
├── Dashboard
├── User Management
│   └── Users · Roles · Permissions
├── Customer Management
├── Driver Management
├── Fleet Management
├── Pricing Management
├── Reports & Analytics
├── Audit Logs
├── Integrations
│   └── Google Maps · ICCES · Notifications · Storage
├── System Settings
├── Profile
└── Settings
```

Shared Platform Services

```
Shared Platform Services
├── Authentication            ├── Media Storage
├── Authorization             ├── Document Management
├── User Roles                ├── Reporting Engine
├── Trip Engine               ├── Audit Logs
├── Pricing Engine            ├── Search & Filters
├── Notification Engine       ├── Activity Logs
├── GPS Tracking Engine       ├── Backup & Recovery
├── ICCES Integration         └── API Services
```

11. Architectural Recommendation — Trip-Centric Architecture

A recommended refinement: make the Operator Web Dashboard the central hub of the entire system. Rather than treating Trips, Fleet, Drivers, and Customers as separate sections, organize everything around the Trip — the platform's core business object, to which every other entity relates.

```
Customer
   │
   ▼
 Trip
 ├── Route                ├── Photos
 ├── Pricing              ├── Timeline
 ├── Driver               ├── Notifications
 ├── Vehicle              └── Reports
 ├── Cargo
 └── GPS Tracking
```

This Trip-Centric Architecture keeps the UI, backend, database, APIs, and reports aligned around the company's primary business process — making the system easier to understand, maintain, and extend as MERCON grows.